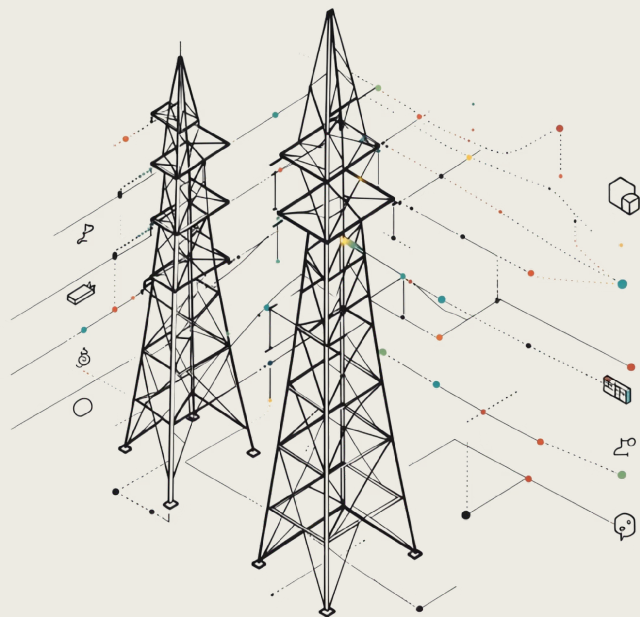




PRAVAHA

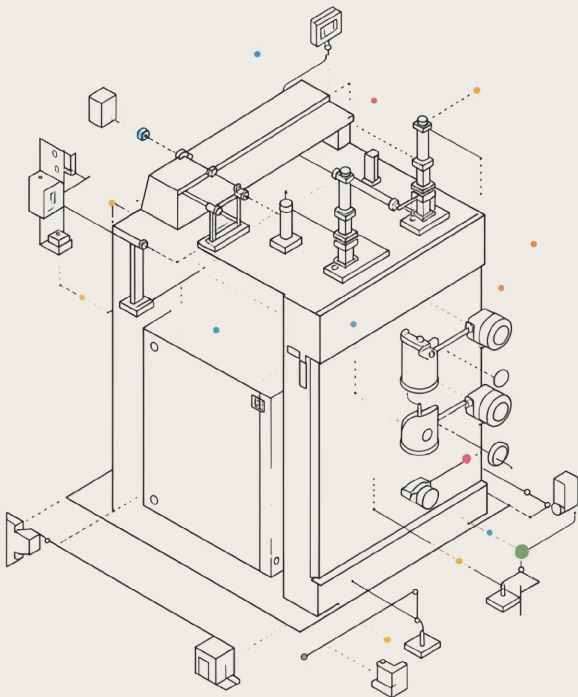
Smarter Insights. Stronger Grids.

AI-powered power-grid risk monitoring and outage-prevention platform for early detection, faster response, and more resilient operations.

Designed to help utilities and operators identify anomalies sooner, prioritize critical assets, and reduce downtime with actionable intelligence.

TEAM VOLTVISION

TRACK: DEVOPS



Power Equipment Can Fail Before Operators Know It

Utilities often have the raw data they need, but it is spread across systems, delayed in reporting, and difficult to interpret before a small issue becomes a costly outage.

The Problem Chain

- **Sensor Readings + Weather + Past Failures**
Transformer and substation data is collected continuously, but it sits in separate tools and formats that are hard to analyze together.
- **Data Is Separated**
Risk is detected too late, and subtle warning signs get missed in raw sensor streams, maintenance logs, and weather signals.
- **Equipment Failure → Power Outage**
Without a way to combine equipment health and weather risk in time, operators cannot act before failures escalate.

Why This Matters

Schedule-Based Maintenance

Traditional maintenance is time-based, not condition-based, so equipment can deteriorate between inspection cycles and fail unexpectedly.

Invisible Warning Signs

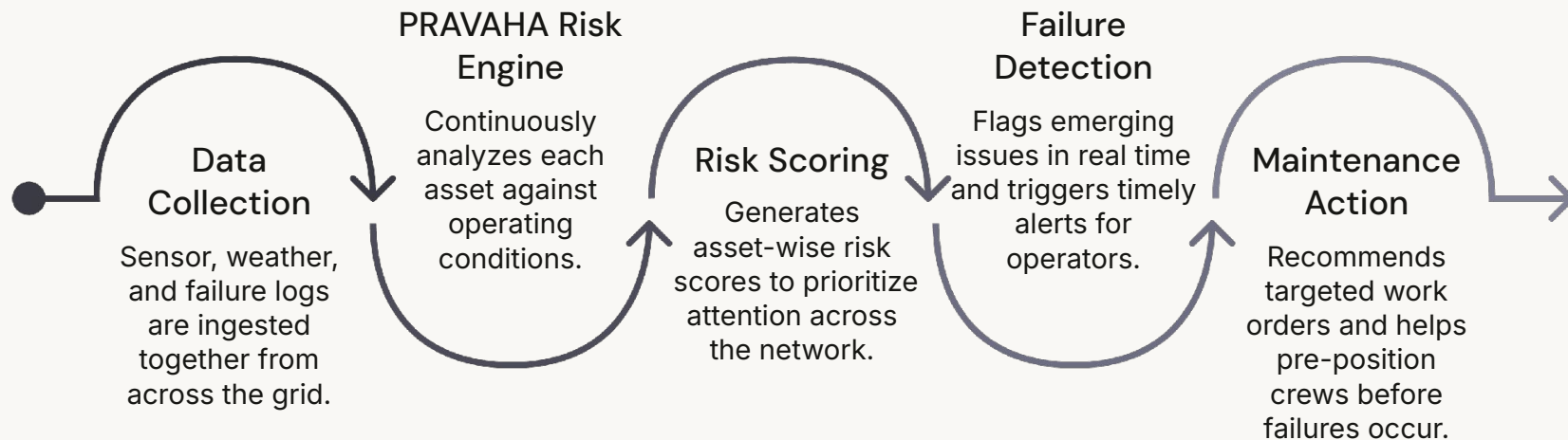
Critical signals may already exist in the data, but without unified analytics they remain buried, unprioritized, and unacted upon.

\$1M+ Per Hour

Transformer and substation failures can trigger outages, emergency response costs, service disruption, and major financial losses in a very short time.

PRAVAHA Turns Grid Data Into Preventive Action

From raw sensor readings to actionable maintenance decisions — in one unified platform that helps utilities detect risk earlier, prioritize the right assets, and act before outages escalate.



 **Know which machine is at risk, understand why it is at risk, and know what to do next.**

Technical Architecture Deep Dive

01

Data Sources & Ingestion

- SCADA sensors capture real-time grid telemetry and operational parameters
- Weather API provides meteorological data and forecasts
- Historical failure logs and maintenance records feed pattern recognition

02

Backend API Layer (FastAPI)

- REST endpoints handle all client requests and data queries
- JWT authentication secures API access
- CORS middleware enables safe cross-origin communication
- ML model inference services integrate predictions into responses

03

Feature Processing Pipeline

- Normalizes telemetry data across different sensor types
- Extracts temporal and statistical features from time-series data
- Combines multi-source signals for comprehensive asset analysis
- Prepares engineered features for ML model consumption

04

ML Inference Engine

- Risk gradient boosting improves prediction accuracy
- Ensemble methods combine multiple model outputs
- Load forecasting predicts future demand patterns
- Real-time scoring generates asset-level risk scores

05

Prediction Service

- Generates risk scores for each monitored asset
- Produces maintenance recommendations based on risk levels
- Outputs crew pre-positioning guidance and priority rankings

06

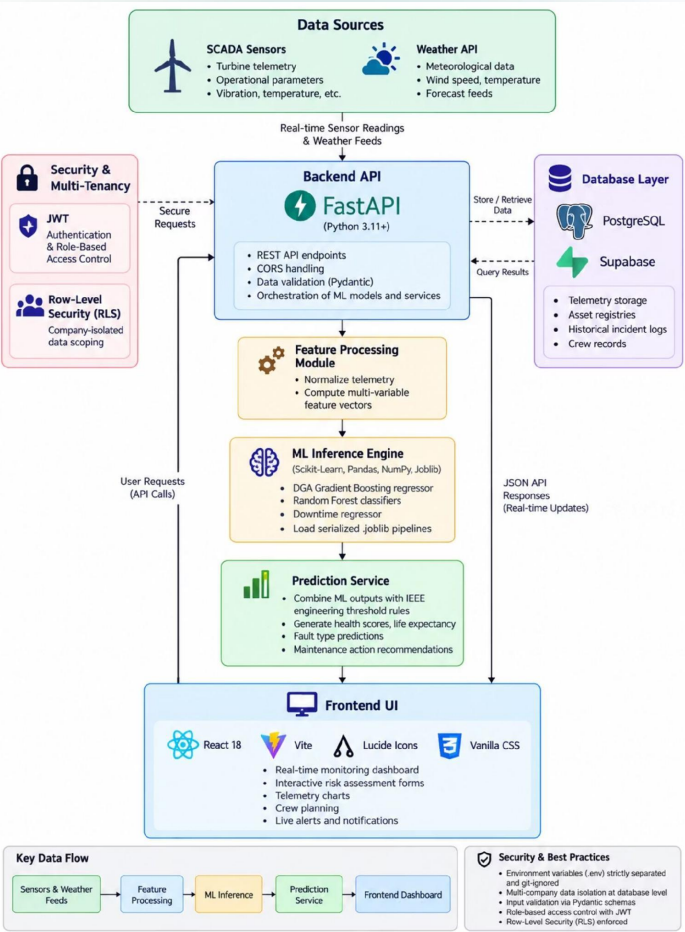
Frontend UI (React + Vite)

- Real-time monitoring dashboard displays live grid status
- Interactive asset detail views show risk breakdowns
- Velocity charts track trends over time
- Alert notifications keep operators informed

07

Database Layer (Supabase PostgreSQL)

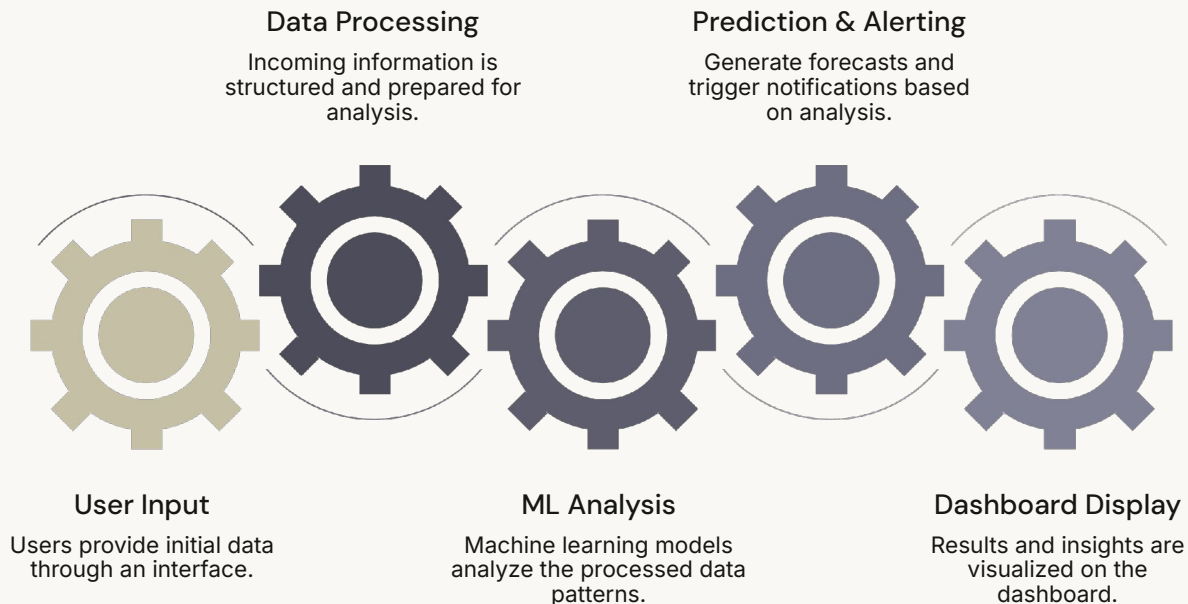
- Normalized schema stores asset metadata and configurations
- Time-series tables optimize sensor reading storage
- Historical incident logs support pattern analysis
- Maintains audit trails for compliance and debugging



How PRAVAHA Works

Full-Stack Architecture

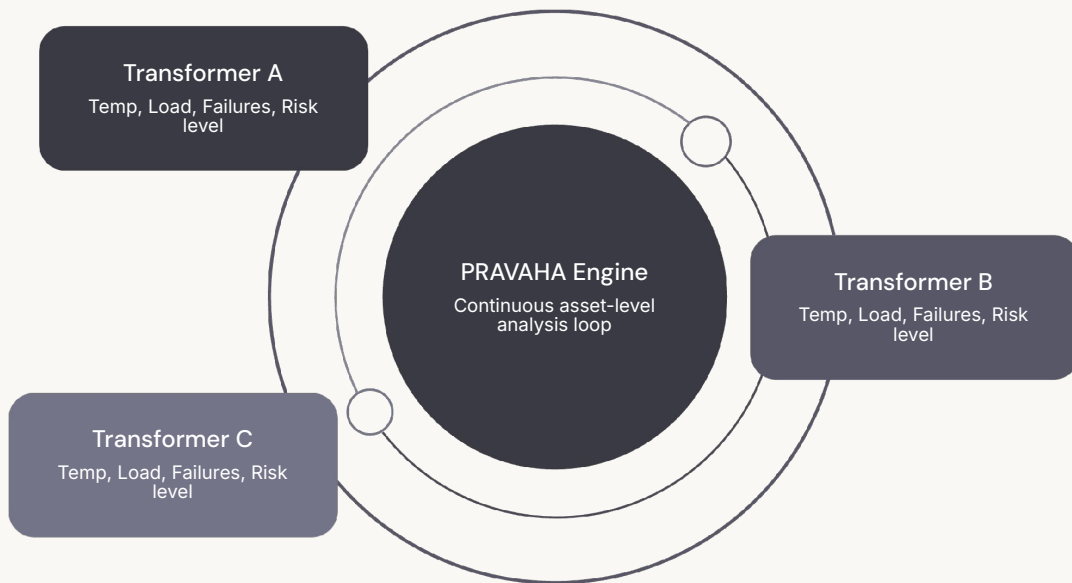
PRAVAHA continuously cycles data through the platform, turning live grid signals into actionable insights and crew guidance.



One Grid. Different Machines. Different Risks.

Asset-Level Intelligence: The Core Advantage

PRAVAHA evaluates each transformer independently, so identical grid conditions can still produce very different outcomes based on the asset's own temperature, loading, and maintenance history.



Different Data → Different Risk → Different Action

That means crews can prioritize intervention where it matters most, instead of treating every asset as if it carries the same level of exposure.

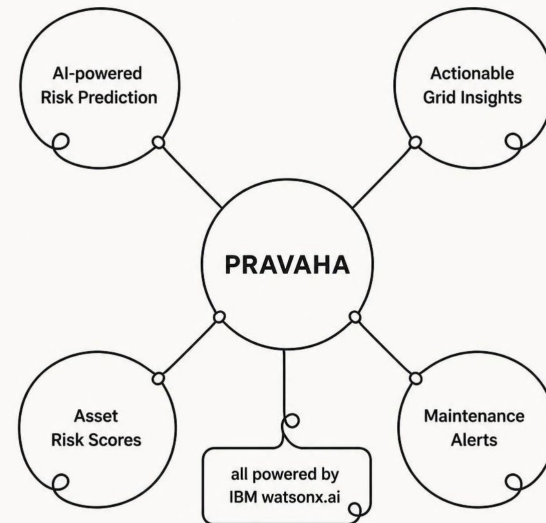
IBM-Powered AI Architecture

IBM watsonx.ai

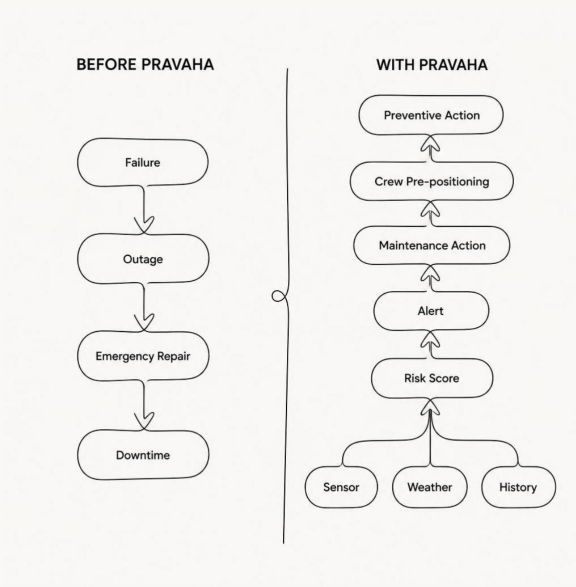
The core AI/ML engine behind PRAVAHA's risk prediction. watsonx.ai processes multi-source grid data — sensor readings, weather conditions, and historical failure patterns — to generate accurate, asset-level risk scores and actionable grid insights. It helps convert raw operational signals into practical intelligence that supports faster, better-informed decisions.

IBM Bob

AI-assisted development and solution building, accelerating PRAVAHA's engineering workflow and enabling rapid iteration on the risk analysis pipeline. It supports prototyping, refinement, and implementation so the team can move from concept to production more efficiently.



From Prediction to Prevention



The PRAVAHA Difference

Earlier Detection

Catch failure signals before they cascade into outages, giving operators more time to intervene and stabilize critical assets.

Smarter Maintenance

Condition-based scheduling replaces costly time-based cycles, so field teams can prioritize work where risk is actually rising.

Faster Response

Crews are pre-positioned and ready before failure occurs, reducing reaction time when conditions begin to deteriorate.

Stronger Grid Reliability

Systematic risk reduction across every monitored asset helps prevent repeat incidents and improves overall service continuity.

Team VoltVision

Meet the team behind PRAVAHA and the roles that drive the project forward. Together, we combine leadership, development, design, integration, and documentation to turn ideas into a practical grid intelligence solution.

Team Member	Role
Arya Shah	Team Lead / Database Analyst / Backend Developer — coordinates the team, manages data structure decisions, and builds the core backend logic.
Bhavishya Ladani	Development / Project Integration — connects modules together, supports implementation, and helps keep the project aligned end to end.
Shakshi Chaudhary	Development / Design — contributes to the system build while shaping the visual and user-facing experience.
Tanvi Monapara	Development / Documentation — supports development tasks and ensures the project is clearly documented for collaboration and future use.

PRAVAHA — Smarter Insights. Stronger Grids.